

Standard Operating Procedure & Troubleshooting Manual

Equipment: Twin-Screw Compounding Extruder Line EX-204 & Melt Discharge System

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Plant Location: Polymer Compounding Facility (Line 2)

Applicability: Process Operators, Maintenance Technicians, Reliability Engineers

1. Equipment Description & System Architecture

Extruder **EX-204** is a 58 mm co-rotating twin-screw extruder driven by a 75 kW AC motor with variable frequency drive (VFD). It is **starve-fed**: gravimetric feeder **F-201** sets the throughput (kg/h), and screw speed sets how full the screws run. The line compounds polypropylene (PP) homopolymer with 20% calcium carbonate (CaCO₃) masterbatch. The discharge section includes a breaker plate, a hydraulic slide-plate screen changer with dual-mesh screen pack (80/120 mesh), and a strand pelletising die.

2. Standard Operating Envelope & Alarm Limits

Table 1: Operating envelope and alarm limits

Sensor Channel	Nominal Operating Range	Warning Limit	Critical Alarm	Emergency Trip / Action
Melt Pressure (Screen Inlet)	140 – 180 bar	190 bar	210 bar	240 bar (rupture disc burst)
Die Melt Temperature	200 – 215 °C	220 °C	225 °C	235 °C (auto heater cut-off)
Motor Drive Torque	50 – 70 %	75 %	80 %	90 % (VFD over-torque trip)
Screw Speed	270 – 290 RPM	± 15 RPM drift	± 25 RPM drift	Feeder cut / interlock
Drive Bearing Vibration	1.5 – 3.0 mm/s	3.5 mm/s	4.5 mm/s	6.0 mm/s (catastrophic stop)
Motor Bearing Temperature	45 – 60 °C	70 °C	75 °C	85 °C (lube failure lockout)
Line Throughput	400 – 500 kg/h	below 380 kg/h	below 350 kg/h	Auto feeder pause

A melt-pressure **rate-of-rise advisory** (ALM-204-P0) is raised when the 3-minute average rise exceeds 2.5 bar/min.

3. Diagnostic Fault Matrix (Troubleshooting Guide)

Table 2: Failure modes, signatures and remediation

Fault ID & Mode	Primary Telemetry Signature	Root Cause Explanation	Prescribed Operator Action
FAULT-A: Screen Pack Clogging (Filter Blinding)	• Melt pressure rises steadily (above 190 bar) • Motor torque rises at constant screw speed • Die melt temperature rises (higher energy input from screws pumping against higher back pressure) • Bearing vibration and temperature remain NORMAL	Un-melted gels, degraded polymer or mineral filler agglomerates block the screen mesh, raising flow resistance.	1. Reduce feeder F-201 rate by 30 % first (throughput drives screen pressure drop). 2. Then reduce screw speed in proportion (about 280 → 200 RPM) to keep the same screw fill. 3. Index the hydraulic screen changer to a clean screen. 4. Ramp feed and speed back together once pressure is below 180 bar.
FAULT-B:	• Bearing vibration rises (above	Loss of lubrication,	1. Stop the motor drive.

Drive Motor Bearing Failure (Mechanical Wear)	3.5 mm/s) • Bearing temperature climbs (above 70 °C) • Melt pressure remains NORMAL • Torque oscillates erratically	bearing raceway spalling, or motor–gearbox shaft misalignment.	2. Do NOT restart under load. 3. Issue urgent mechanical work order.
FAULT-C: Feed Throat Bridging (Polymer Starvation)	• Melt pressure drops sharply (below 90 bar) • Motor torque drops (below 35 %) • Output surges and drops • Melt temperature drops	Premature pellet softening in the hopper throat due to loss of throat cooling water.	1. Inspect throat cooling water flow. 2. Dislodge bridge using brass rod. 3. Verify feeder operation.
FAULT-D: Polymer Thermal Degradation (Overheating)	• Melt temperature above 225 °C across zones • Melt pressure decreases (lower viscosity) • Black specks/char in strands • Acrid degradation odour	Thermocouple malfunction, barrel heater runaway, or excessive screw shear rate.	1. Check barrel zone thermocouples. 2. Purge line with high-MFI HDPE. 3. Lower barrel zone setpoints by 15 °C.

4. Incident Escalation Protocol

When SCADA issues concurrent alarms across pressure, torque and temperature, the lead operator must first differentiate between:

- **Flow-resistance faults (Fault-A):** high pressure + high torque + normal vibration and bearing temperature.
- **Drive mechanical faults (Fault-B):** high vibration + high bearing temperature + normal pressure.

On a starve-fed extruder, **never reduce screw speed alone** at full feed rate: the screws run fuller and torque can rise further. Reduce feed first.

Never restart an overloaded extruder without checking screen pack differential pressure.

When a raw-material lot change precedes a process upset, quarantine the lot and notify QA.